

A Short Introduction to Vector Search and RAG

A primer for apprentices. This document is intentionally factual and self-contained so you can use it as a test fixture: ask your app questions about it and check the answers against the text.

What is an embedding?

An **embedding** is a list of numbers (a vector) that represents the meaning of a piece of text. Texts with similar meaning get vectors that are close together. In this project, embeddings are produced by a local model served through Ollama. The **nomic-embed-text** model outputs a vector with **768 dimensions** for each piece of text.

Measuring similarity

To find text similar to a question, we compare vectors using **cosine similarity**, which measures the angle between two vectors. Cosine similarity ranges from **-1 to 1**: a value near 1 means the texts are very similar in meaning, and a value near 0 means they are unrelated.

Chunking

Long documents are split into smaller pieces called **chunks** before embedding. Chunks that are too large dilute the meaning and retrieve irrelevant text; chunks that are too small lose context. A common starting range is **200 to 800 tokens** per chunk, often with a small overlap between consecutive chunks so a sentence is not split awkwardly across a boundary.

Retrieval: top-k

When a user asks a question, the question is embedded and compared against every stored chunk. The system keeps the **top-k** most similar chunks (for example, the top 4) and passes them to the language model as context. Choosing k is a trade-off: too few chunks may miss the answer, too many add noise and cost.

Why this reduces hallucination

Retrieval-Augmented Generation (RAG) combines retrieval with generation. Instead of answering from memory, the model is given the retrieved chunks and asked to answer using only that context, citing its sources. If no relevant chunk is found, a well-built RAG app should respond that it does not know rather than inventing an answer. This grounding in real source text is what makes the output trustworthy.